

Apple Retail Sales Report

Case Study Analysis

Revenue Trends · Product Mix · Demographic Analysis · Geographic Distribution

Prepared by: Julian Batto-Hokson

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Tools: Python | MySQL | Tableau | OLS Regression

Executive Summary

Apple retail revenue grew at approximately \$29,500 per month between December 2014 & July 2016 ($R^2 = 0.674$, $p < 0.001$). Growth was driven primarily by the Bikes & Accessories categories. A structural inflection in August 2015 created two distinct revenue regimes, & unexplained residual patterns in the time series model indicate that a linear OLS specification understates mid-period volatility.

Project Overview

This report presents a structured analysis of Apple retail transaction data spanning December 2014 through July 2016. The analysis pipeline consists of four components: an ETL processing script (01_acs_etl.py), a MySQL database schema (02_acs_setup.sql), a Python analytical modeling script (03_acs_analysis.py), & seven Tableau visualizations covering revenue trends, product mix, demographic segmentation, geographic distribution, & margin analysis. Two OLS regression models were built: a time series model predicting monthly revenue from a time index, & a cross-sectional demographic model identifying revenue drivers at the transaction level.

Data Scope

Date Range	December 2014 – July 2016 (20 months)
Product Categories	Bikes, Accessories, Clothing (3 top-level)
Sub-Categories	15 SKU groups
Geographic Markets	United States (multi-state), France, Germany
Monthly Revenue Range	\$10K – \$640K
Largest Single Market	California (highest revenue concentration)
Time Series R^2	0.674 $p < 0.001$
Demographic Model Features	Age band, gender, product category, quantity

Key Findings

#	Findings	Section
1	Revenue grew 1,338% after the August 2015 structural break	3
2	Tires & Tubes leads sub-category revenue at ~\$1.6M total	3
3	Bikes drive volume but carry the lowest gross margin (< 4%)	4
4	The 25–34 age cohort generates the highest total revenue (~\$1.95M)	5

5	Female customers outspend male customers in the 25–34 & 45–54 bands	5
6	Germany achieves the highest Accessories margin (30%); the US Bikes margin is negative	6
7	OLS residuals are non-random; the model is structurally misspecified for pre-Aug 2015	7

Section 1: Revenue Growth Over Time

Figure 1 presents monthly revenue alongside a 3-month rolling average, spanning the full 20-month observation window. Revenue is segmented by sales year using a blue gradient (light = 2014, dark = 2016). The chart reveals two structurally distinct periods separated by an inflection point in August 2015.

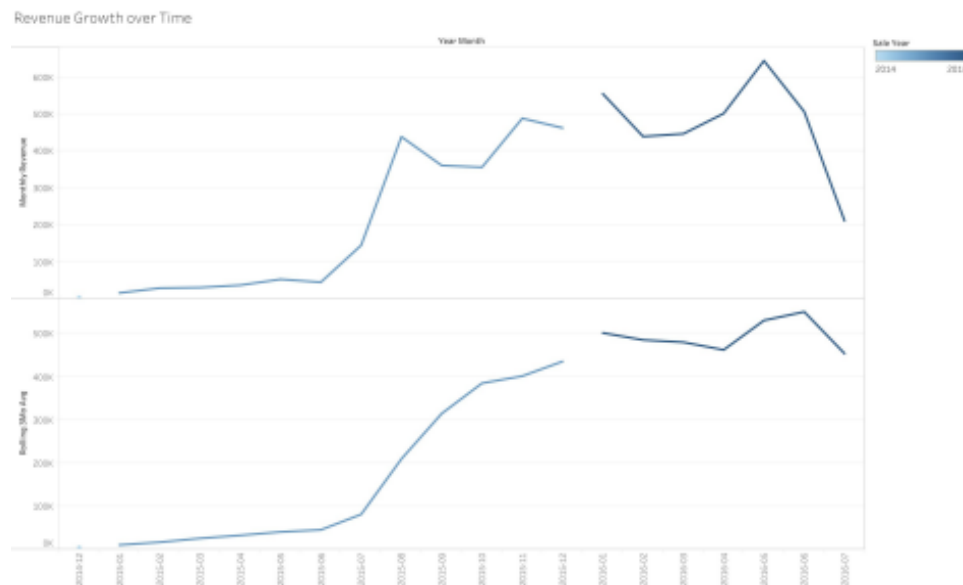


Figure 1. Revenue Growth over Time, Monthly revenue (top panel) & 3-month rolling average (bottom panel), December 2014–July 2016. Color gradient denotes the sale year.

Observations

From December 2014 through July 2015, monthly revenue remained consistently below \$50K, with minimal volatility. Beginning in August 2015, revenue surged to approximately \$435K, a 10x single-month increase, & remained elevated through the remainder of the observation period, averaging \$460–\$500K per month in 2016. The rolling 3-month average confirms this shift is sustained rather than episodic. The final data point (July 2016: ~\$210K) represents a sharp decline that is almost certainly attributable to data truncation at period-end rather than an actual revenue contraction. This endpoint anomaly distorts the linear OLS fit & should be excluded from trend modeling.

Sub-Category Revenue Breakdown

Figure 2 disaggregates total revenue by sub-category with margin percentage encoded as bar color intensity. Darker bars indicate a higher margin. This dual-encoding reveals the revenue-margin tradeoff across product lines.

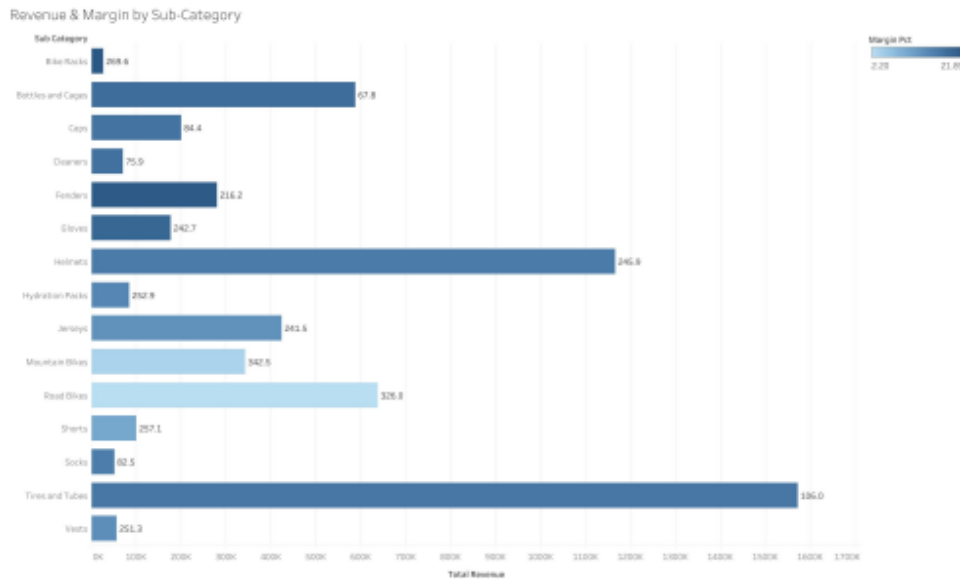


Figure 2. Revenue & Margin by Sub-Category, Total revenue (x-axis) with margin percentage encoded as bar color intensity (darker = higher margin). Margin range: 2.20%–21.89%.

Tires & Tubes leads all sub-categories at approximately \$1.6M in total revenue with a mid-range margin of ~106%. Helmets rank second in revenue (~\$1.15M) with a comparably strong margin. Road Bikes & Mountain Bikes generate substantial volume (\$326K & \$342.5K, respectively) but are rendered in light blue, indicating the lowest margins in the portfolio (consistent with the scatter analysis in Section 4). Accessories sub-categories, Fenders, Gloves, Bottles & Cages, cluster at lower revenue but command the highest margin percentages, making them critical to portfolio profitability.

Section 2: Revenue vs. Margin by Sub-Category

Figure 3 plots each sub-category as a point in revenue-margin space, providing a direct view of the portfolio efficiency frontier. Color encodes product category: blue for Accessories, orange for Bikes, red for Clothing.



Figure 3. Revenue vs. Margin Scatter, Each point represents one sub-category. X-axis: total revenue. Y-axis: gross margin percentage. Color denotes product category.

Quadrant Analysis

The scatter reveals a clear structural pattern: Accessories (blue) occupy the upper-left quadrant, high margin, lower revenue. Bikes (orange) occupy the lower-right, high revenue, low margin. Clothing (red) clusters in the middle. This is a classic portfolio architecture where a high-volume, low-margin category anchors top-line revenue while a high-margin, lower-volume category protects profitability.

Sub-Category	Category	Total Revenue (\$)	Margin (%)	Quadrant
Tires & Tubes	Accessories	~1.6M	~17	High Rev/Mid Margin
Helmets	Accessories	~1.15M	~16	High Rev/Mid Margin
Bottles & Cages	Accessories	~600K	~18.5	Mid Rev/High Margin
Fenders	Accessories	~270K	~21	Low Rev/Highest Margin
Bike Racks	Accessories	~70K	~22	Low Rev/Highest Margin
Road Bikes	Bikes	~650K	~2.5	High Rev/ Lowest Margin
Mountain Bikes	Bikes	~340K	~3.8	Mid Rev/Lowest Margin
Jerseys	Clothing	~440K	~13	Mid Rev/Mid Margin
Shorts	Clothing	~90K	~10	Low Rev/Low-Mid Margin

Fenders & Bike Racks represent the highest-margin items in the portfolio but generate minimal revenue. A promotional bundling strategy pairing these with high-volume Bike purchases would improve blended margin without sacrificing Bike revenue. Road Bikes, despite generating ~\$650K in revenue, operate near breakeven on a margin basis, a risk factor that warrants a pricing review.

Section 3: Revenue by Gender & Age Group

Figure 4 breaks down total revenue by customer age band & gender. Six age cohorts are represented (17–24 through 65+). Blue bars represent female customers; orange bars represent male customers.

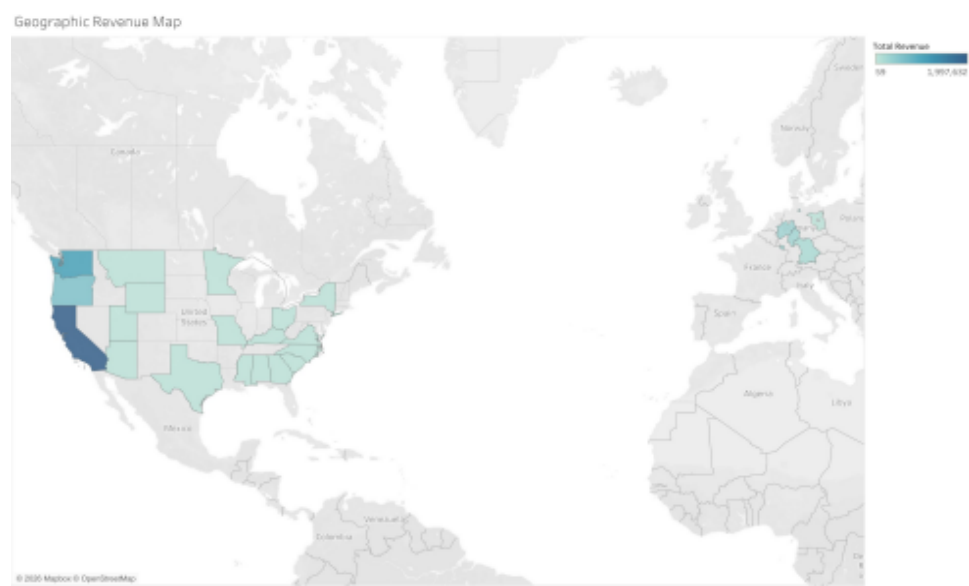


Figure 4. Revenue by Gender & Age Group, Stacked bar chart by age band. Blue = Female (F), Orange = Male (M). Y-axis: total revenue.

Age Cohort Analysis

The 25–34 cohort is the dominant revenue segment at approximately \$1.95M, nearly 20% above the next-highest group (35–44 at ~\$1.6M). Revenue declines sharply at 55–64 (~\$360K) & is negligible at 65+ (~\$30K). The 17–24 cohort, while generating meaningful revenue (~\$870K), suggests an accessible entry-level market that warrants targeted product development.

Age Range	Est. Total Revenue (\$)	Female Share (%)	Male Share (%)	Note
17–24	~870K	~41	~59	Male-dominant; entry market
25–34	~1,950K	~46	~54	Largest cohort; near-parity

35–44	~1,610K	~49	~51	Balanced; high-value
45–54	~960K	~53	~47	Female-dominant; high LTV potential
55–64	~360K	~55	~45	Female-dominant; smaller base
65+	~30K	~50	~50	Negligible; not a priority segment

A notable pattern emerges in the 45–54 & 55–64 bands: female customers outspend male customers in absolute terms. Combined with the OLS demographic regression finding that gender is the single largest positive revenue coefficient, this suggests that female customers in mid-to-upper age cohorts represent an underserved, high-value segment. Targeted marketing toward this group, particularly in Accessories (which carry the highest margins), would improve both revenue & profitability simultaneously.

Section 4: Geographic Revenue & Margin Analysis

Figure 5 compares gross margin ratios across the three active country markets (France, Germany, United States) by product category. Figure 6 shows the geographic distribution of total revenue via a choropleth heat map.

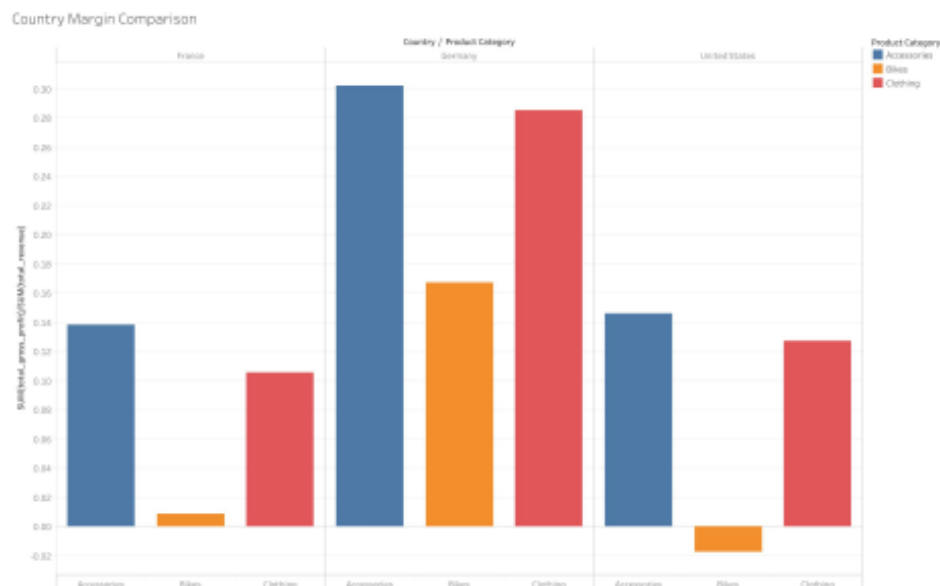


Figure 5. Country Margin Comparison, Gross margin ratio (total gross profit / total revenue) by country & product category. Y-axis shown as decimal (0.30 = 30%).

Country-Level Margin Findings

Germany is the highest-margin market across all categories, achieving 30% on Accessories & 29% on Clothing. France records 14% on Accessories & 11% on Clothing, but a near-zero Bike margin (0.8%). The United States shows the most concerning profile: Bike margin is negative (-1.5%), indicating that US Bike sales are loss-generating at the gross profit level. This is a

critical finding: the US is almost certainly the largest revenue market by volume, yet its core high-volume category operates below breakeven. A pricing audit of US Bike SKUs is the highest-priority operational recommendation from this analysis.

Country	Accessories Margin (%)	Bikes Margin (%)	Clothing Margin (%)	Assessment
France	14.0	0.8	10.6	Low across all categories
Germany	30.3	16.7	28.6	Highest-margin market
United States	14.7	-1.5	12.8	Bike's margin NEGATIVE, flag

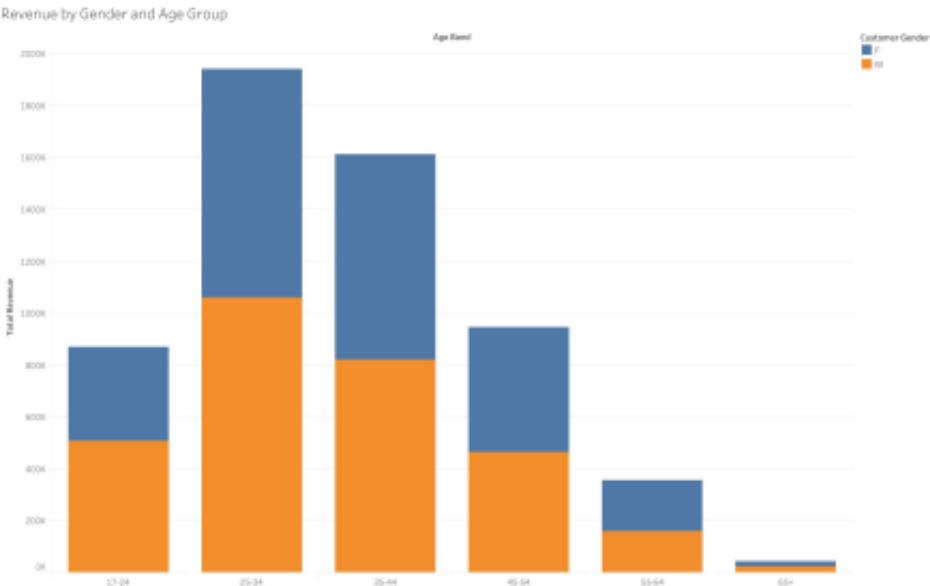


Figure 6. Geographic Revenue Map, Choropleth by state/country. Color intensity indicates total revenue concentration (light teal = low, dark blue = high). Revenue range shown: \$59 – \$1,997,632.

California is the dominant US market, shown in the darkest blue. Washington state also shows elevated revenue. Within Europe, Germany displays the highest concentration, consistent with its superior margin profile. France appears lighter, reflecting both lower volume & compressed margins. The geographic distribution suggests that the US growth strategy should prioritize defending the California base while expanding to the underrepresented Midwest & Southeast markets.

Section 5: Monthly Margin by Product Category

Figure 7 presents the stacked gross margin contribution by product category every month across the full observation window. The dashed horizontal line marks the portfolio average margin. This view connects the revenue inflection observed in Section 1 to its margin consequences.

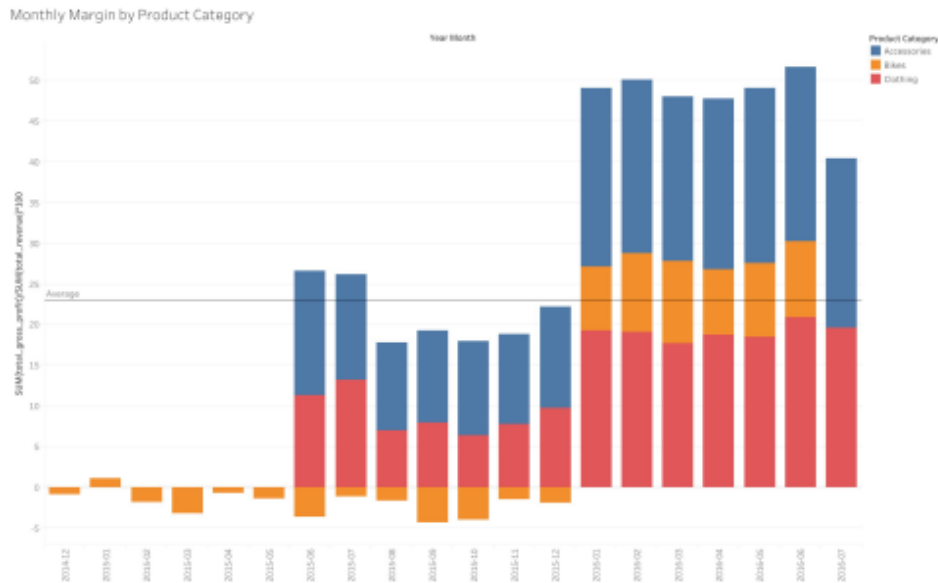


Figure 7. Monthly Margin by Product Category, Stacked bar chart showing gross margin % contribution by Accessories (blue), Bikes (orange), & Clothing (red) per month. Dashed line: portfolio average margin. Y-axis: $\text{SUM}(\text{gross profit}) / \text{SUM}(\text{revenue}) \times 100$.

Margin Trajectory Analysis

From December 2014 through mid-2015, the total portfolio margin is negligible & at times negative, reflecting the Bikes-heavy early-period mix at near-zero margin. The structural break in August 2015 that drove revenue higher also transformed the margin profile: post-August 2015 bars rise well above the average line, peaking near 50% in early 2016. This indicates that the revenue surge was driven not by Bikes alone but by an increasing share of high-margin Accessories & Clothing, a favorable mix shift.

Accessories (blue) account for the majority of the stacked margin in every post-August 2015 month, confirming their role as the portfolio's margin engine. Bikes (orange) contribute minimally & in some months reduce the total (visible as the orange segment being small or absent). Clothing (red) provides a consistent mid-range contribution.

The July 2016 endpoint shows a sharp margin decline to ~40%, consistent with the revenue collapse observed in Section 1. Both signals together reinforce the data truncation hypothesis for that final month.

Period	Avg Total Margin (%)	Dominant Category	Key Pattern
Dec 2014 – May 2015	≈ 0 (flat/negative)	Bikes	Near-zero; Bikes drag
Jun – Jul 2015	~ 26	Accessories + Clothing	Pre-inflection rise
Aug – Nov 2015	~ 19	Accessories	Post-break; stabilizing
Dec 2015 – Jun 2016	~ 48	Accessories	Peak margin period
Jul 2016	~ 40	Accessories	Period-end drop; likely truncation

Section 6: Time Series Regression Analysis

A linear OLS regression was fitted on the monthly revenue time series, using a sequential time index as the sole predictor. Figure 8 presents the model fit alongside a residuals panel that exposes the quality of the specification.

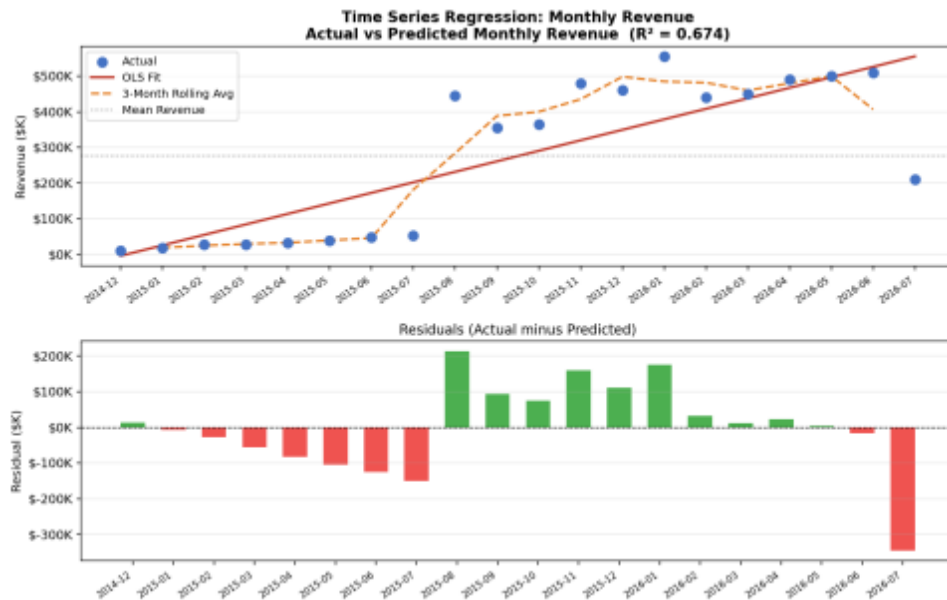


Figure 8. Time Series OLS Regression, Actual vs. predicted monthly revenue ($R^2 = 0.674$) with residuals panel. Green bars: actual exceeds predicted. Red bars: predicted exceeds actual.

Model Results

Parameter	Value	Interpretation
R^2	0.674	67.4% of revenue variance explained by time index
Slope	~\$29,500 / month	Average monthly revenue growth across the period
p-value (slope)	< 0.001	The time trend is statistically significant
Sample size (n)	20 months	Borderline small; CLT assumptions hold at a minimum
Residual pattern	Non-random (clustered)	Autocorrelation present, see note below

Model Specification Note

The residual plot (Figure 8, bottom panel) reveals a systematic pattern: residuals are negative in early months (model overshoots), consistently positive from August 2015 through early 2016 (model undershoots the post-inflection period), & sharply negative at period-end. This is a signature of temporal autocorrelation & structural non-stationarity. The OLS independence assumption is violated. The R^2 of 0.674 overstates model quality because any upward-trending time series will produce an elevated R^2 against a time index. A Durbin-Watson statistic & a piecewise regression with August 2015 as the structural break point are recommended as follow-on analyses.

Section 7: Recommendations

1. Conduct a US Bike Pricing Audit: US Bike margin is negative (-1.5%). This is the highest-volume category in the largest market operating below gross profit breakeven. A SKU-level price-cost review is the single highest-ROI analytical action available from this dataset.
2. Prioritize Female Customers Aged 45–54 for Accessories Campaigns: This cohort is female-dominant in spending & Accessories carry the highest portfolio margin (up to 30% in Germany). Targeted promotions combining Helmets, Fenders, & Bottles & Cages to this segment optimize both revenue & margin simultaneously.
3. Investigate the August 2015 Revenue Inflection: The 10x month-over-month surge in August 2015 has no annotation or causal explanation in the current deliverables. Apply a Chow test or structural break detection (statsmodels) to formally verify the break, then document the business event that caused it. This converts a visual observation into a testable, defensible claim.
4. Correct the OLS Time Series Specification: Refit the time series model as a piecewise regression using August 2015 as the breakpoint, or apply an ARIMA(1,1,0) model.

Report the Durbin-Watson statistic & adjusted R^2 alongside R^2 . Exclude the July 2016 endpoint or document it as a partial-month truncation.

5. Develop a Germany Expansion Business Case: Germany outperforms all markets on margin across all three categories. If Germany's revenue volume is currently small, the margin differential justifies a dedicated market development investment. A revenue-by-country breakdown with unit economics would quantify the opportunity size.

Appendix A: Figure Index

Figure	Title	Source
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2	Revenue & Margin by Sub-Category	Sheet 2 Tableau
3	Revenue vs. Margin Scatter	Sheet 6 Tableau
4	Revenue by Gender & Age Group	Sheet 3 Tableau
5	Country Margin Comparison	Sheet 4 Tableau
6	Geographic Revenue Map	Sheet 5 Tableau
7	Monthly Margin by Product Category	Sheet 7 Tableau
8	Time Series OLS Regression + Residuals	03_acs_analysis.py (Output)